

SPIE Photonics West – Presentation Summaries

Juan David Muñoz-Bolaños · Medical University of Innsbruck · April 2026

Presentation 1 — Photonics West 2025 (BiOS)

19 March 2025 · SPIE Proceedings Vol. 13328 · Adaptive Optics and Wavefront Control for Biological Systems XI · DOI: 10.1117/12.3039528 [cite:1]

Title: Wavefront corrections over large fields of view via beam cone tomography [cite:1]

Authors: Juan D. Muñoz-Bolaños, Johannes Locher, Maria Borozdova, Kibum Nam, Simon Moser, Monika Ritsch-Marte, Alexander Jesacher [cite:1]

When imaging objects embedded in scattering tissue, light gets distorted in complex ways. These distortions, called aberrations, are not uniform and change from point to point across the image. Standard adaptive optics methods usually correct aberrations only at a single point, so the correction weakens when moving to a different region of the image. [cite:1]

This presentation describes a tomographic approach that reconstructs a 3D map of refractive index variations inside the scattering sample. From that map, it becomes possible to estimate wavefront distortions at different positions across a larger field of view. The reconstruction uses wavefront measurements from several test points around the target area and was validated with numerical simulations and proof-of-concept experiments. [cite:1]

In practical terms, the work shows how a spatial aberration map can support wavefront correction over a much larger area than standard single-point adaptive optics methods. [cite:1]

Presentation 2 — Photonics West 2026 (OPTO)

5 March 2026 · SPIE Proceedings Vol. PC13918 · Emerging Digital Micromirror Device-based Systems and Applications XVIII · DOI: 10.1117/12.3076734 [cite:1]

Title: Scatter compensation for enabling deep two-photon microscopy using a MEMS phase light modulator [cite:1]

Authors: Juan David Muñoz-Bolaños, Simon Moser, Florian Harrasser, Monika Ritsch-Marte, Alexander Jesacher [cite:1]

Two-photon microscopy can image deep inside biological tissue, but increasing depth also increases scattering and reduces image quality. Adaptive optics can compensate for this, yet many correction devices are too slow for live and dynamic biological samples. [cite:1]

This presentation introduces a fast and lower-cost solution based on a MEMS phase light modulator from Texas Instruments. The device is combined with DASH, a wavefront-shaping method designed for scatter correction that benefits from high-speed modulation. The

reported result is a signal improvement of 1 to 2 orders of magnitude, corresponding to images that are about 10 to 100 times brighter and clearer at depth. [cite:1]

The presentation argues that this MEMS-based approach can make real-time scatter correction more practical for deep-tissue multiphoton microscopy while reducing dependence on slower and more expensive modulators. [cite:1]